

Martin (2023) Single-Instrument Futures Momentum

\$50k capital · EWMAC 6 speeds equal-weight · per-trade skew characterisation

As of
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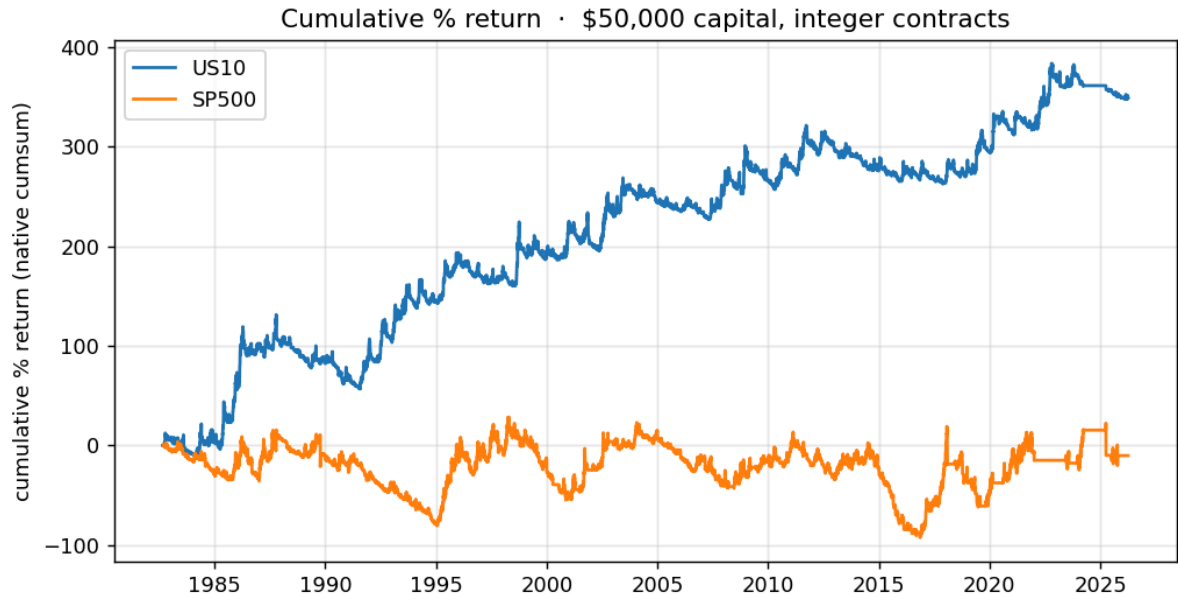
Headline Verification

- PASS** US10 (rates) shows positive daily skew [Martin Fig1] *skew_daily = +0.176*
- PASS** SP500 (equity) shows negative daily skew [Martin Fig1] *skew_daily = -1.988*
- PASS** Per-trade skew >> daily skew on US10 [Martin: trend injects positive skew at trade level] *skew_per_trade = +4.38 vs daily +0.18*
- PASS** Low win rate + strong positive trade skew [Martin: convex / long-option payoff] *win% = 23.4 · skew_per_trade = +4.38*
- PASS** US10 gross Sharpe in [0.2, 0.6] [refutes 0.7-1.0 claim, matches Carver/Martin] *Sharpe_gross = 0.396*

Summary Metrics

instrument	ann_ret_g %	ann_vol %	Sharpe_g	Sharpe_n	skew_daily	maxDD %
US10	8.873	22.431	0.396	0.35	0.176	-74.543
SP500	-0.112	24.534	-0.005	-0.009	-1.988	-120.874

Equity Curves



Native cumsum %-return curves of single-instrument 6-speed EWMAC (equal-weight, carry off, vol-target 20%, soft cap 20, integer contracts). US10 (rates) compounds steadily; SP500 (equity) is flat to negative, consistent with Martin Fig 1 sign contrast.

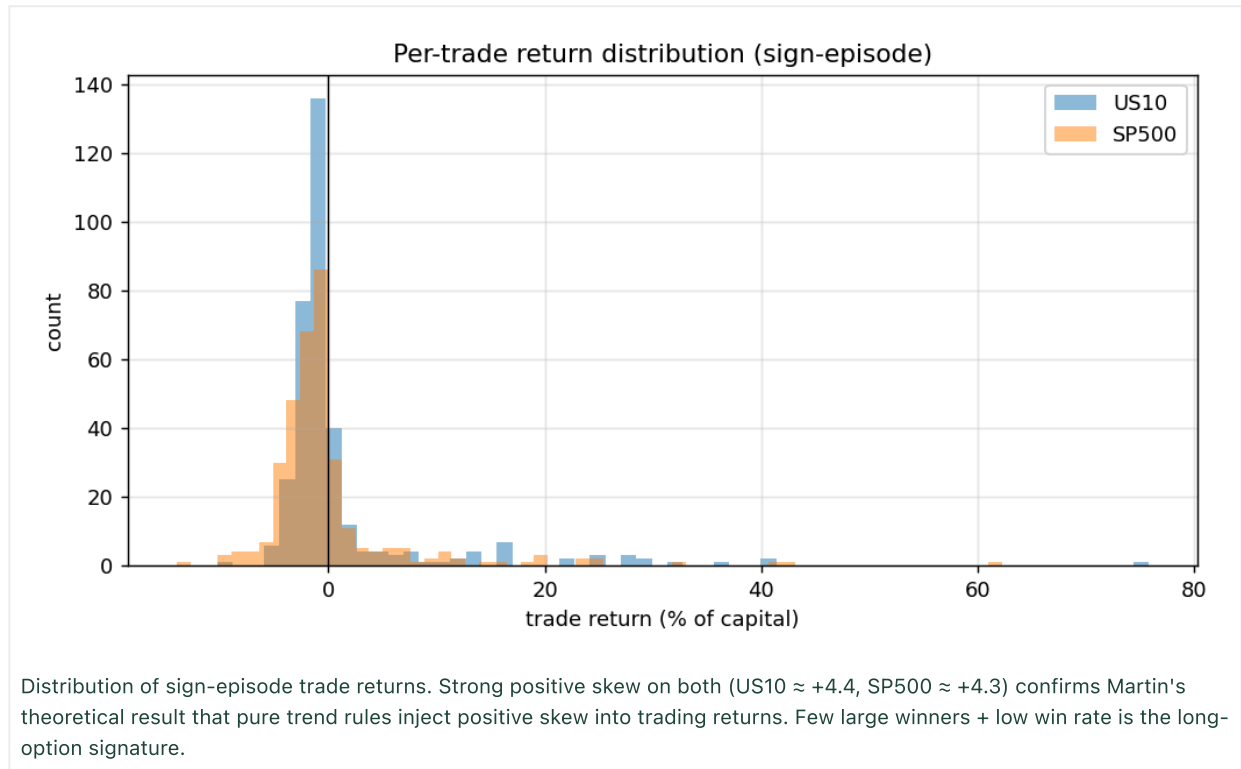
Literature: Richard J. Martin, "Design and analysis of momentum trading strategies" (2023). See references/research/.

Per-Trade Analysis

Sign-episode trade ledger · US10, SP500

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Per-Trade Distribution



Trade Statistics

instrument	n_trades	win %	med days	med ret %	skew_trade	turn/yr
US10	342	23.4	12	-1.116	4.382	34.41
SP500	335	19.7	12	-1.306	4.327	33.47

Run Manifest

Engine	native pystemtrade futures_chapter15 (do_not_reimplement_engine=true)
Instruments	US10, SP500
Trading rules	ewmac2_8, ewmac4_16, ewmac8_32, ewmac16_64, ewmac32_128, ewmac64_256
Forecast weights	equal (1/6 each), carry off
Forecast cap (soft)	20
Vol target (%)	20
Capital (USD)	50,000
Round positions	1
IDM	1
Trade definition	sign-episode: maximal run of constant non-zero sign(rounded buffered position)
Python	3.10.15

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Arki Robust Standard · do_not_reimplement_engine=true

Run 20260528T151806Z · git c3494ee2

